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Paleomagnetism of Lower Devonian volcanics and Devonian dykes from northcentral New Brunswick, Canada

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Some 50 oriented samples (120 specimens) have been collected on eight sites of volcanic rocks from the Lower Devonian Dalhousie Group of northern New Brunswick and Devonian andesitic to basic dykes from central New Brunswick. Univectorial and occasional multivectorial components were extracted from the various samples. Results after AF and thermal demagnetization compare relatively well. In the volcanics and tuffs, two components of magnetization have been isolated: A ($D = 33°$, $I = -58°$, $\alpha_{95} = 7.3°$, $K = 236$) for four sites and B ($D = 66°$, $I = +53°$) for three sites. The grouping of component A is improved after tilt correction but the fold test is not significantly positive at the 95% confidence level. Component A is interpreted as being primary while component B is unresolved and appears to be the resultant magnetization of a Late Paleozoic and a recent component. The pole position obtained for tilt corrected component A is 268°E, 1°S, $d_p = 6.5°$, $d_m = 8.8°$. The paleolatitude calculated for component A is 39°S. The paleopole of *in situ* component A is located close to those of the Early–Middle Devonian formations from Quebec, New Brunswick and New England states while the paleopole of tilt-corrected component A is similar to Lower Devonian poles of rock units from the Canadian Arctic Archipelago. If component A is primary (as we believe it to be), then the western half of the northern Appalachians had already docked onto the North American Craton by Early Devonian time. Alternatively, if component A is secondary the same conclusion applies but the juxtaposition took place in Middle Devonian time.

1. Introduction

Recent literature on Devonian paleomagnetic results of North America (Dankers, 1982; Johnson and Van der Voo, 1983; Spariosu and Kent, 1983; Roy et al., 1983) show that the data set is not only limited in number to make paleozoic continental restorations but also suffers from several ambiguities. The amount of scatter in published pole positions for the Siluro–Devonian, particularly with regard to intrusive rocks, seems to preclude any discussion of the APW path for the Devonian period or of relative motion of displaced terrains during the Acadian orogeny. Considering the existing state of the paleomagnetic data, the construction of a Devonian APW path is hypothetical and debatable but may become realistic as new reliable paleomagnetic data are being published. The present paleomagnetic study is from the Lower Devonian volcanic units (Dalhousie Group) of northern New Brunswick and Devonian diabase dykes from central New Brunswick.

2. Geological setting

The Chaleurs Bay and Dalhousie Groups in northern New Brunswick consist of several volcanic units interlayered with generally fine clastics commonly containing a fauna of Late Silurian and Early Devonian age. These lithological units rest unconformably upon the Ordovician rocks

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(Tetagouche and Elmtree Groups) of the Miramichi anticlinorium and have been deformed during the Devonian Acadian orogeny into concentric folds cut by steep to vertical faults (Poole and Rogers, 1972).

The Dalhousie Group consists of a sequence of shallow water fossiliferous shale and limestone and intercalated beds of andesite and basalt flows and tuffs. The underlying calcareous shale may be Early Devonian or Upper Silurian. The tuff occurs as local pockets and lenses at various levels within the Sunnyside Formation in northern New Brunswick. The basic flows probably originated from vents at Sugar and Dalhousie Mountains near Campbellton and Dalhousie.

The Dalhousie Group, sampled on sites D-3 to D-7, was studied by Gesner (1843), Logan et al., (1863), Hind (1865), Honeyman (1878) and Ells (1881). Clarke (1909) provided the first detailed account of the stratigraphy and paleontology of the Dalhousie section, establishing it as the best representative shelly fauna of the Lower Devonian Helderbergian stage (Gedinnian) outside of its type area in the State of New York.

Howard (1926) studied the igneous rocks of the Dalhousie area and attempted to relate them to different volcanic centres (Fig. 1). Although only two series of flows and tuffs were found east of Dalhousie Mountain, no fewer than five similar series were traced toward the west. Of these, the second is the most nearly continuous and extends for over 20 km west of Dalhousie (Howard, 1926). The Sugar Loaf dacites, which are abundant in the vicinity of Campbellton, are gray porphyric rocks with phenocrysts of oligoclase or andesine, with some hornblende and quartz in a groundmass of feldspar and quartz. Magnetite and apatite are constant accessories.

Felsic tuffs containing plant fragments occur at the top of the section at Campbellton. These tuffs

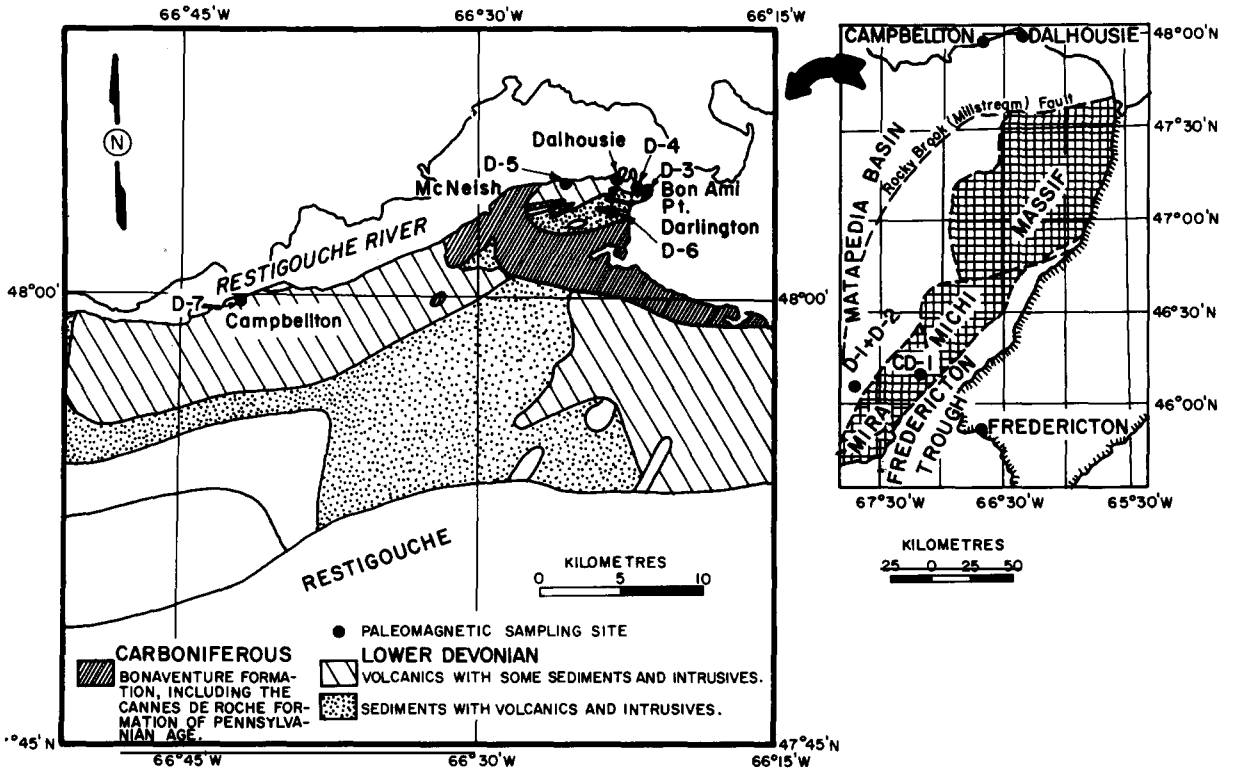


Fig. 1. Simplified geologic map of the Dalhousie–Campbellton region, northern New Brunswick and of the west-central New Brunswick region with the distribution of paleomagnetic sampling sites as indicated by numbers next to solid circles in the vicinity of Campbellton, Dalhousie and Fredericton.

are unconformably overlain by fish-bearing continental sediments of the early Middle Devonian Campbellton Formation (Dineley and Williams, 1968; Greiner, 1974). The tuffs associated with dacite flows are massive in structure and composed of quartz, orthoclase, oligoclase, andesite, magnetite and a very fine grained, almost glassy base. Samples D-3, D-4, and D-5 were taken from andesitic tuffs near the top of the section at Dalhousie while sample D-6 is from the base of the section. Sample D-7 includes both basaltic and felsic tuffs near the top of the section at Campbellton.

The Gibson granodiorite (Anderson, 1968; Venugopal, 1979), located just south of Woodstock in west-central New Brunswick, is a post-Acadian pluton of Early Devonian age (Fyffe et al., 1981). The oval-shaped stock, with a surface area of ~ 15 km^2, possesses a narrow border phase of fine-grained, light grey porphyritic granodiorite intruded by a core of pinkish grey, medium- to coarse-grained hornblende–biotite granodiorite.

Two narrow diabase dykes (sites D-1 and D-2) intruding the granodiorite on Highway 105 on the east side of the Saint John River, were sampled. The diabase consists of sparce phenocrysts of plagioclase set in a matrix of flow-aligned plagioclase laths. Original mafic minerals are completely replaced by chlorite. Vesicles are filled with epidote, calcite, chlorite and quartz. The Lower Devonian Becaquimec Lake Gabbro, lying about 30 km northeast of Woodstock in west-central New Brunswick, forms a northerly trending oblong pluton with a surface area of ~ 150 km^2. It is composed principally of medium- to coarse-grained ophitic gabbro; troctolite is abundant in the southern portion. Small stocks and dykes of porphyritic hornblende–biotite granite intrude the gabbro (Anderson, 1968; Venugopal, 1982).

Mafic dykes, representing the youngest intrusive phase present in the Becaguimec Lake Gabbro, were sampled for paleomagnetic study (Sample CD-1). The dykes are fine-grained diabases typically containing narrow, very fine-grained porphyritic trap dykes at their contacts with the medium-grained gabbro of the main phase of the pluton.

In the present study, oriented hand samples at five sites (26 samples) from volcanics and tuffs of the Dalhousie Group and three sites (15 samples) from diabase dikes were collected near Campbellton and Dalhousie and in central New Brunswick (Fig. 1).

3. Measurements and results

The remanent magnetization was measured with a Schonstedt digital spinner magnetometer (DSM-1). The volume susceptibility was measured at room temperature in the Earth's field ($H = 0.05$ mT) using a Cenco MS-3B susceptibility bridge. Stepwise demagnetization tests were carried out using a Schonstedt thermal demagnetizer (TSD-1) with residual fields of < 15 nT in the cooling chamber, and an alternating field (AF) demagnetizer, with a 3-axis tumbler, in about 5 nT residual fields (Seguin, 1975). Stepwise AF and thermal procedures were employed up to 100 mT and 600°C and Zijderveld (1967) vector diagrams were used in part of the subsequent analysis. For the NRM of each oriented sample, at least two specimens (diameter 2.5 cm, height 2.2 cm) were used to obtain a sample mean direction. NRM directions and intensities of all the oriented samples of volcanics, tuffs and diabase dikes are shown in Fig. 2. Approximately 25% of these samples have a NRM direction close to the present Earth's field (PEF) at $D = 340°$ and $I = 75°$. The remaining 75% has scattered directions away from the PEF, mainly concentrated in the northeastern quadrant. The remanence of most specimens that lie close to the PEF were found to move away from the initial directions during subsequent AF or thermal treatment. The NRM intensities (J_n) range from 3×10^{-3} to 9×10^{-1} Am^{-1} with values of Koenigsberger ratio ($Q_n = J_n/KH$) between 0.2 and 2 (mean = 0.8).

3.1. Volcanics and tuffs

One specimen from each of 26 samples (five sites) was demagnetized in alternating fields up to 100 mT. Most samples lost 50% of their NRM intensity by 30 mT and 90% by 85 mT. In some specimens a large fraction of the NRM intensity

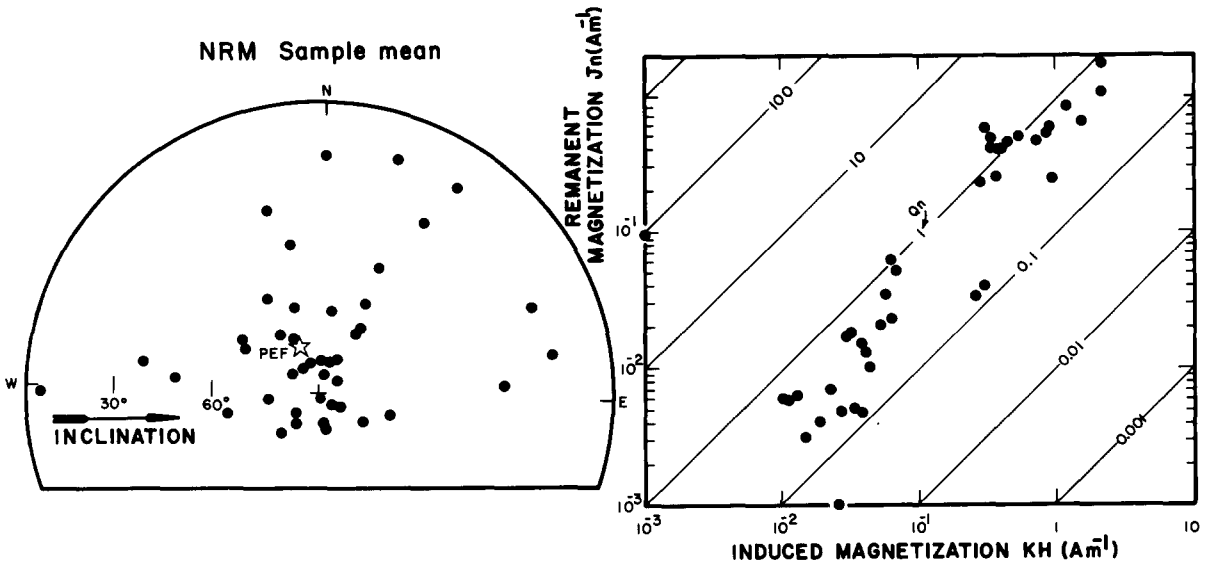
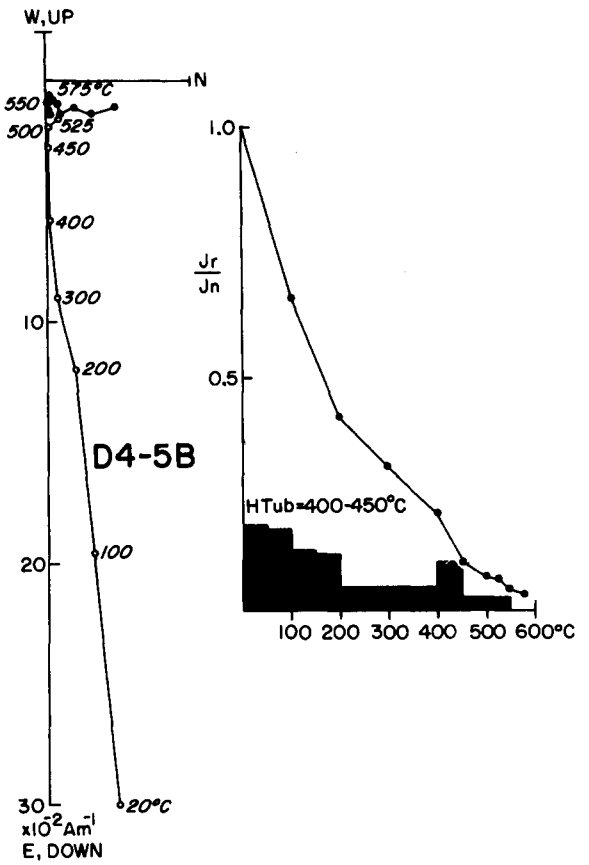


Fig. 2. Sample NRM directions plotted on Wulff net for Dalhousie Group volcanics and tuffs as well as diabase dykes (left). Solid circles represent projections on the lower hemisphere. The star represents the present Earth's field (PEF) of the sampling region. Plots of corresponding intensity of NRM (J_n in Am^{-1}) against the induced magnetisation (KH in Am^{-1}) with resulting slope lines indicating the values of Koenigsberger ratio, (Q_n right).



still remains at 100 mT. Most of the samples have a straight decay to the origin of the magnetization vector above 25 mT which may mean occurrence of only one component of magnetization. Two components of magnetization were isolated by AF treatment in some samples (e.g., D-4). Component A has a SSW shallow to intermediate downward direction and component B a ENE intermediate downward direction. Figure 3 shows a typical example of the behavior of the B component upon thermal demagnetization. The A component was isolated in the 15–100 mT range but most commonly between 30 and 60 mT. Figure 4 shows a standard to sub-standard example of the behavior of the A component in the course of demagnetization. Figure 5 shows a typical example of difficult extraction of the A component (normal and reverse) through AF demagnetization. Figure 5a indicates a Zijderveld's diagram (sample D5 – 5A)

Fig. 3. Representative orthogonal thermal demagnetisation diagram (Zijderveld, 1967) of successive end-points of magnetisation vector for specimen D4-5B from the volcanics. The solid (open) circles represent projections in the horizontal (vertical) plane. Numbers next to open circles represent the demagnetisation steps (in °C). Representative normalized intensity ((J_r/J_n) decay curves and unblocking temperature for the same specimen in the bottom part of the figure.

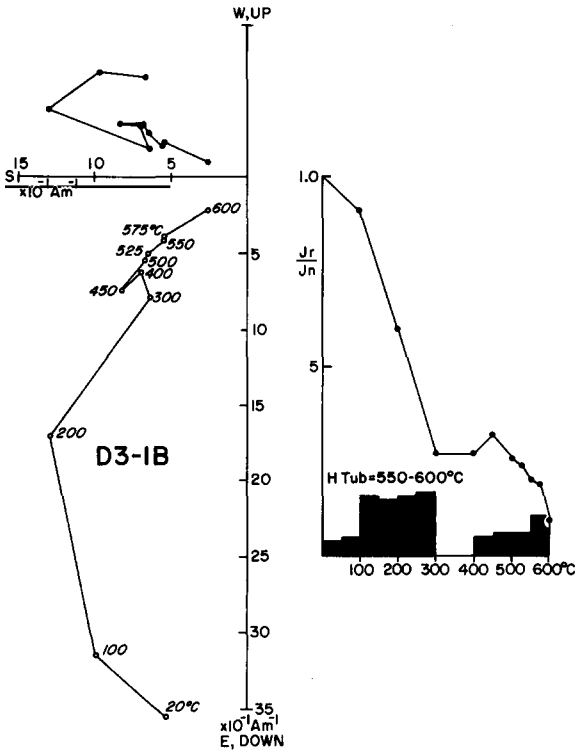


Fig. 4. Representative demagnetisation diagram for specimen D3-1B. Same descriptions as in Fig. 3.

characterising component A (normal). Figure 5b illustrates a Zijderveld's diagram (sample D3-2) characterising component A (reverse). After removal of a WSW component with steep downward inclination (+60 to +80) in the 0–30 or 40 mT range, a single residual component (straight lines on the Zijderveld's diagram) is usually obtained in the 30 or 40–100 mT range. This residual SW component with shallow to intermediate downward inclination is most often univectorial (straight lines on the Zijderveld's diagram); occasionally, the projected vectors do not end at the origin.

Component B was usually isolated in the low field (c. 10 mT) range. Component A is more frequent than component B; it is found in most of the samples of four sites (D-3, D-4, D-5 and D-7) whereas component B is present in three samples of site D-4 only. The marked linear spread of the directions of site D-4 is largely due to the presence of two components of magnetization (A and B).

One to two specimens of each sample of the five volcanics and tuffs sites were subjected to thermal treatment in 100°C steps up to 400°C and in closer temperature steps up to 650°C until they showed a linear decay towards the origin of the vector diagrams (Fig. 6). Figure 6b illustrates the typical example of the A (R) (reverse) component in the course of thermal treatment. A steep inclination is removed in the 20–300°C temperature range. Above this temperature, a SSW shallow to intermediate downward direction is the only component remaining; the endpoint is reached at ~ 575–600°C. Figure 6a shows another example of an A (normal) component specimen. Most commonly, a univectorial component (straight lines) is isolated in the 400–600°C range. The projected vectors do not reach the origin in all instances.

Two components of magnetization (A and B) were also isolated with thermal treatment. In almost all the cases, the AF demagnetised and thermally cleaned components of the samples coincided fairly well (Table I). Most samples lost 50% of their NRM intensity by 350°C and 80% by 570°C. Very commonly, the unblocking temperature (T_{UB}) of the A and B components are approximately the same (T_{UB}) range: 450–575°C. As for sites D-3, D-5 and D-7, site D-6 is characterised by a univectorial component but the direction of this component (W, intermediate upward) is different from component A. The D-6 component is best isolated in the 15–60 mT and 525–575°C ranges. On site D-6, a very rapid intensity decrease takes place within a very narrow range (570–610°C). Because of its uncertain stratigraphic position and its higher degree of alteration the paleomagnetic results of this site are not included in the final statistics of the A and B components.

3.2. Diabase dykes

As for the volcanics and tuffs, 40% of the specimens were AF demagnetised and 60% thermally. Most of the samples lost 50% of their NRM intensity by 12 mT and 250°C and 90% by 35 mT and 550°C. Only one component (B) was isolated on the samples of sites D1 and D2. It is characterised by a NE intermediate to steep downward direction, a median destructive field of 5–10 mT

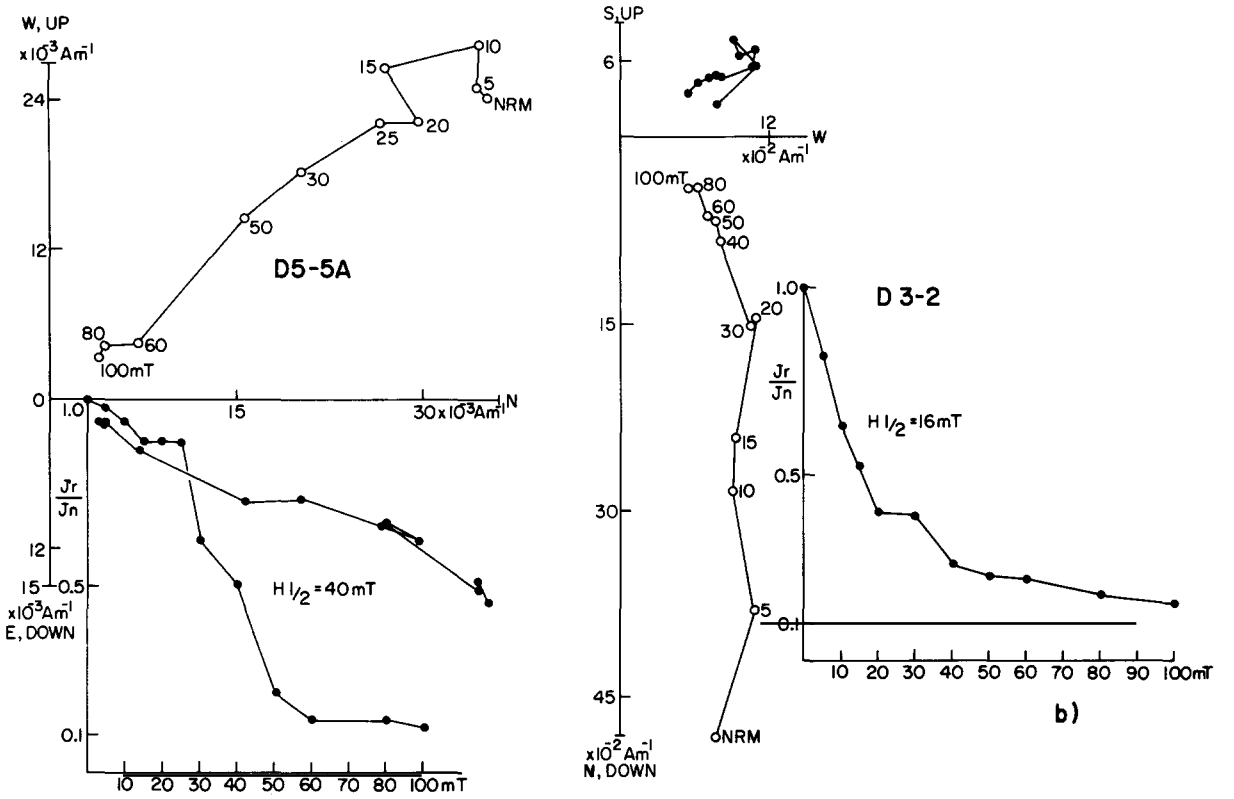


Fig. 5. Representative orthogonal AF demagnetisation diagrams (Zijderveld, 1967) of successive end-points of magnetisation vector for two specimens (D5-5A and D3-2) from the volcanics. The solid (open) circles represent projections in horizontal (vertical) plane. Numbers next to open circles represent the demagnetisation steps (in mT). Representative normalised intensity (J_r/J_n) decay curves and median destructive field for the same two specimens are indicated in the bottom part of the figure.

and a T_{UB} of 300–575°C. A possible component of magnetisation (SE, steep downwards) was identified in three specimens of site D-1. We believe that there are too few specimens on only one site to constitute a separate component. Thus, only one component of magnetisation was isolated on diabase dykes D1 and D2. Figure 7 shows Zijderveld's diagrams displaying the behavior of component B during AF demagnetization. In Fig. 7a the component is represented as a single univectorial component decaying down to the origin. In Fig. 7b, the situation is complicated by the superposition of a NNW steep downward component (PEF?) in the 0–20 mT range. The residual univectorial B component is isolated in the 20–100 mT range, but the intersection of the origin is not always obtained.

Figure 8 displays a typical vector diagram for the unresolved component B. In the 100–450°C range, the NE intermediate to steep downward B direction is well isolated. From 500 to 600°C, an additional component (NNW steep downward, PEF?) is obtained. It is our opinion that "component B" from sites D1 and D2 is an unresolved component. The arguments for this statement are the following: (1) occasionally, a NNW steep downward component (PEF) is extracted at higher temperatures, in a field free space (15 mT); (2) quite frequently, the projected vectors do not reach the origin on the diagrams; and (3) the intensity of magnetisation decreases very rapidly with an increase of alternating field strength and temperature. Thus, component B appears to be a mixture of older (Paleozoic) and younger (Recent) mag-

TABLE I

Demagnetized NRM results for all individual specimens

Site No:	No:	n'/n	*D* (°)	*I* (°)	AF intensity (mT)	*T* (°C)
D-1	1-A		54	57	20, 30, 40	
	1-B		67	66		525
	1-C		67	64		400, 450
	2		66	53	30, 40, 50, 60	
	3-A		51	65		550, 575
	3-B		145	63 *	20, 30, 40, 60	
	3-C		136	70 *		525, 550, 575
	4-A		53	52		400, 450
			125	60 *		525
	4-B		55	65	20, 30, 40, 80, 100	
	4-C		45	63		200. 400
	5-A	11/13	43	51	50	
UNIVECTORIAL COMPONENT						
D-2	1-A		53	58	50	
	1-B		57	62		400, 600
	1-C		60	72		400
	2-A		77	42	10–30	
	2-B		51	47		500
	2-C		81	47		450
	3-A		45	59		500
	3-B		44	63	30–40	
	4-A		65	64	30 – 40	
	5-A		95	31		300–500
	5-B		88	24		300–500
	5-C	12/16	76	23	40–60	
UNIVECTORIAL COMPONENT						
D-3	1-A		205	42	60–100	
	1-B		202	38		550–570
	2		235	36	60	
	3-A		205	40	80–100	
	3-B		205	42		575–600
	4		296	64 *	50–60	
	5-A		224	40		550–575
	5-B		233	42	100	
	5-C	9/9	200	35		525
UNIVECTORIAL COMPONENT						
D-4	1-A		221	34		500–525
	1-B		199	36	60–100	
	2-A		62	54	10	
			20	– 27	80–100	
	2-B		70	56		525–550
	3-A		177	32	5–20	
	3-B		167	37		300–500
	4-A		211	36	15–30	
	4-B		78	46		500
	5-B	9/13	65	52		525
TWO COMPONENTS						
D-5	1-A		20	– 35	25–30	
	1-B		33	– 40		500–525
	2-A		24	– 37	10–15	
	2-B		27	– 38		525–575
	2-C		28	– 39		450
	3-A		320	– 38 *	30–50	
	3-B		88	– 65 *		450–525

TABLE I (continued)

Site No:	No:	n′/n	D (°)	I (°)	AF intensity (mT)	T (°C)
	3-C		132	−55 *		400–550
	4-A		13	−35	40–45	
	4-B		48	−31		565–575
	5-A		30	−39	40–50	
	5-B		31	−32		500–550
	6-A		23	−46	10–60	
	6-B	14/18	27	−31		525–550
UNIVECTORIAL COMPONENT						
D-6	1-A		292	−60	15–60	
	2-A		287	−68	40–50	
	2-B		281	−75		550–575
	3-A		268	−56	40–50	
	3-B		274	−56		550
	3-C		274	−57		550
	4-A		306	−57	15–40	
	4-B		326	−57		525–550
	5-A		285	−63	60	
	5-B	10/12	279	−56		525–550
UNIVECTORIAL COMPONENT						
D-7	1-A		8	−60	60	
	1-B		5	−65		600–625
	1-C		4	−61		525–550
	2-A		32	−40	8–100	
	2-B		38	−38		575–600
	2-C		33	−32		550–575
	3-A		358	−50	50–80	
	3-B		359	−51		600–650
	4-A		23	−40	80–100	
	4-B		16	−42		450–575
	4-C		20	−34		525–550
	5-A	12/13	24	−25		575
UNIVECTORIAL COMPONENT						
CD-1	1-A		251	−37		600
	1-B		270	−33	20–50	
	2-A		282	64		575
	2-B		299	28	20–30	
			272	70	40	
	2-C		280	62	30	
			266	29	50	
	3-A		293	61	40–60	
	3-B		299	22		600
	3-C		175	72		535–600
	3-D		200	68		570–600
	4-A		316	−29		400–575
	4-B		16	44 *	30–50	
	4-C		49	01 *		530–600
	4-D		75	70 *	80–100	
	5-A		304	06		500
	5-B		325	08		600
	5-C	16/18	320	17	100	
MULTIVECTORIAL COMPONENTS						

$n'/n = \frac{\text{number of demagnetized specimens}}{\text{total number of specimens}}$.

* specimens excluded from the statistics.

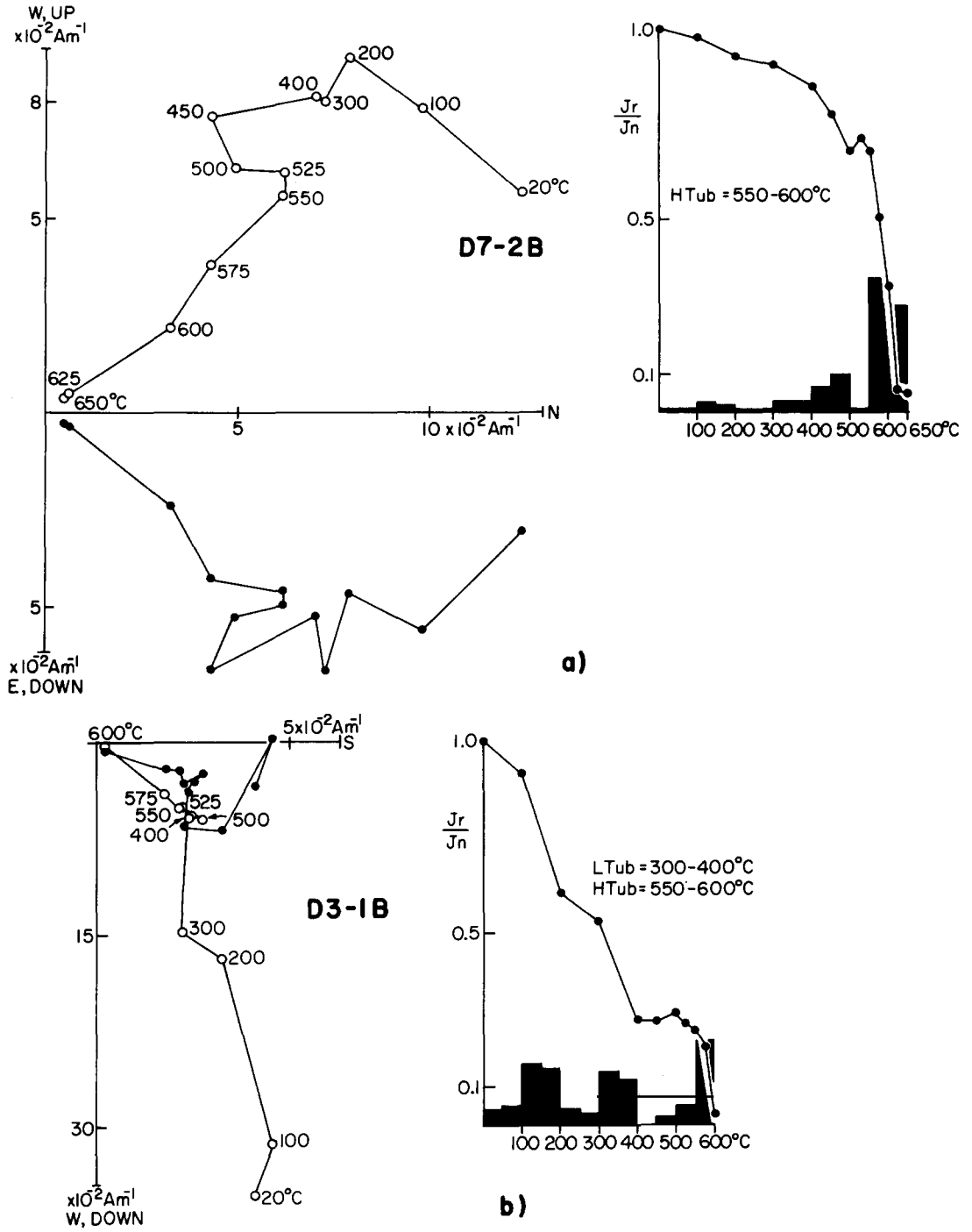


Fig. 6. Representative orthogonal thermal demagnetisation diagrams (Zijderveld, 1967) of successive end-points of magnetisation vector for two specimens (D7-2B and D3-1B) from the volcanics. Numbers next to open circles represent the demagnetisation steps (in °C). Representative unblocking temperatures (T_{UB}) spectrum after stepwise thermal demagnetisation for the same two specimens as indicated in the bottom part of the figure. The other specifications are the same as for Fig. 5.

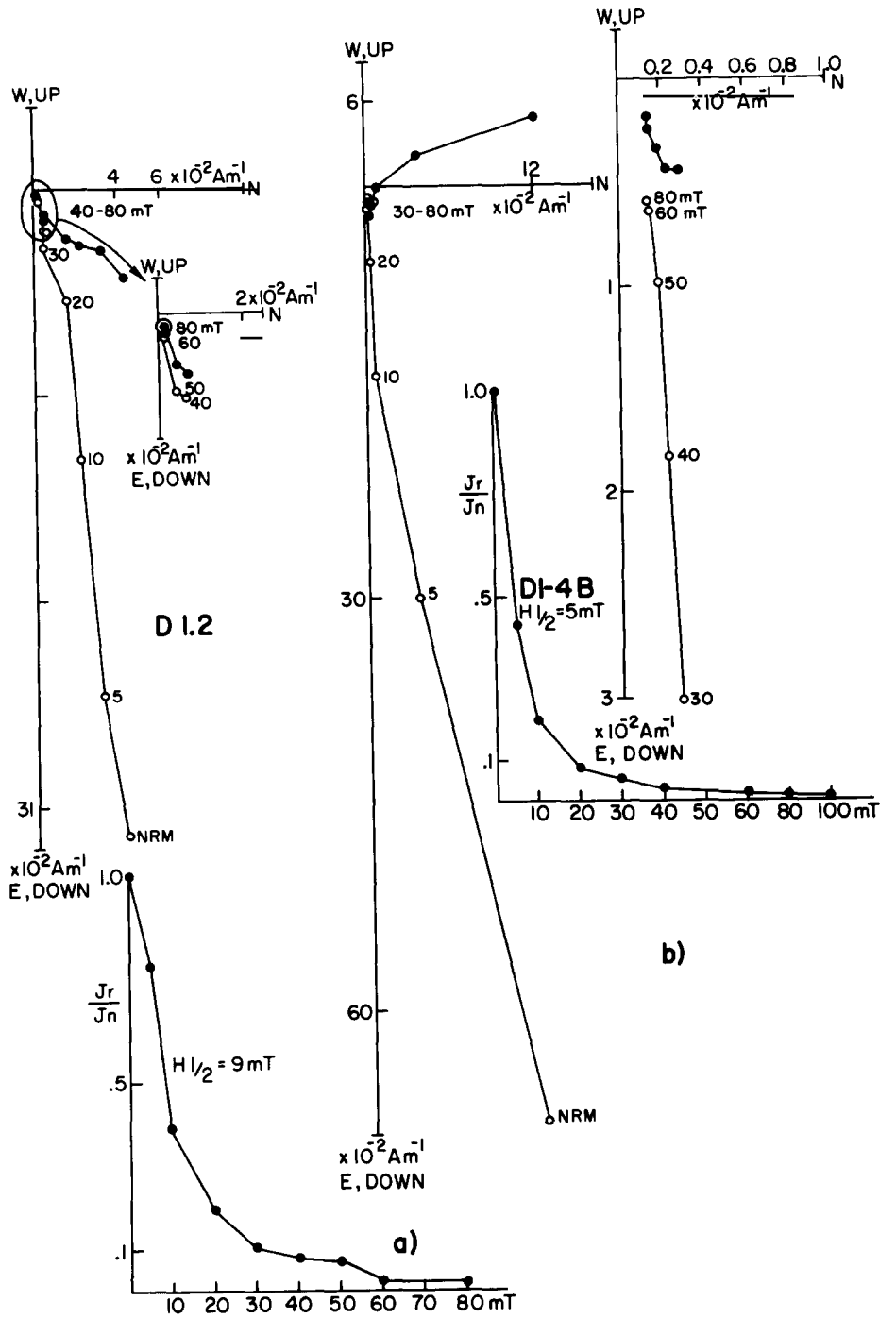


Fig. 7. Representative orthogonal AF demagnetisation diagrams (Zijderveld, 1967) of successive end-points of magnetisation vector for two specimens of diabase (D1-2 and D1-4B). For the other descriptions, see Fig. 5.

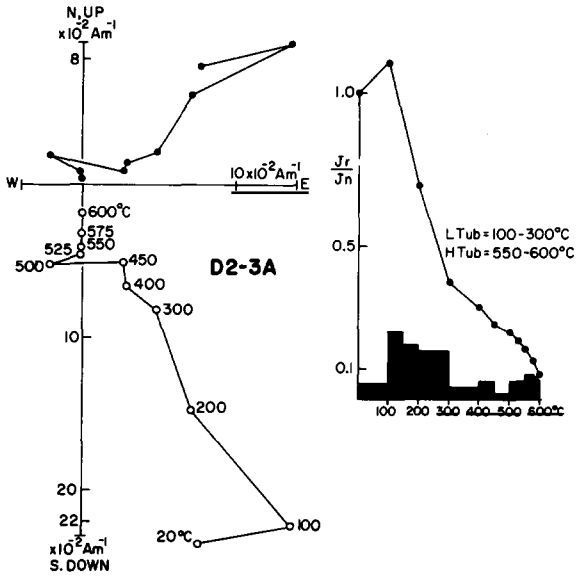


Fig. 8. Representative orthogonal thermal demagnetisation diagram (Zijderveld, 1967) of successive end-points of magnetisation vector for one specimen of diabase (D2-3A). For additional descriptions, see Fig. 5.

netisations. B magnetisation may not be a true paleomagnetic component.

Site CD-1 which consists of a mixture of diabase and trap dykes is characterised by multiple components of magnetisation. This intrusion of traps within the diabase is indicative of time varying, crustal extension, implying extended past tectonic activity. It is noted that site CD-1 is located in the Miramichi massif (Fig. 1) where recent earthquakes (1982) have occurred. Thus, the above mentioned observations suggest that Devonian tectonic activity continued sporadically up to the present time. These geological observations suggest that CD-1 multicomponents correspond to different ages of magnetisation, and for this reason this site is not pertinent to this interpretation of Devonian paleomagnetic data. No mention of the results from this site will be made in the rest of this paper.

4. Analysis of paleomagnetic directions

The site-mean directions of magnetisation were computed and classified into two groups (A and B) based on the similarity in directions for the volcanics, tuffs and diabase dykes (Table II).

4.1. *Volcanics and tuffs*

Based on criteria such as stable end-points, magnitude of within-site scatter and to a lesser extent range of T_{UB} and coercivity spectra, two groups of magnetisation directions are recognized. The A group is a southwesterly directed remanence with shallow to intermediate positive (downward) inclination. It characterises the Dalhousie volcanics and tuffs because it was observed at four of the five sites and was obtained from samples with stable end-points. Mixed (two normal and two reverse sites) polarities are observed in this group. The A group-mean direction *in situ* is 26°, −37° and 33°, −58° after tilt correction. The B group is a ENE directed remanence (which may be unresolved) with intermediate to steep positive (downward) inclination. It was observed on 1 site (D-4) only. Tests based on stability criteria such as Graham's (1949) fold test can be performed to infer the age of magnetisation of group A with respect to the time of deformation. A comparison of the estimate of Fisher's (1953) precision parameter between *in situ* and tilt-corrected directions suggests a tilt test which is neither negative nor significantly positive at the 95% confidence level for this A component (McElhinny, 1964). However, the geological dips are almost uniform and relatively small ($\simeq 25°$); consequently it is difficult to conclude whether the magnetism is primary or secondary. The A component appears to be the characteristic remanence of the volcanics and tuffs in view of its frequent occurrence as well as its higher H_C and occasional higher T_{UB} values; the higher coercivity and T_{UB} values of the A component are mainly due to the presence of fine-grained magnetite and hematite. Mineralographic observations suggest that hematite is primary and not an oxidation product of magnetite. Consequently, the group-mean direction of the tilt-corrected A component most likely represents the oldest component in the volcanics and tuffs but it is difficult to tell whether it is primary or secondary.

TABLE II

Site-mean remanence directions after AF and thermal treatments combined

Site No.	N	n	D_o	I_o	α_{95}	K_0	D_1	I_1	K_1
	In-situ						After tilt correction		
Volcanics and tuffs									
Group A									
D-3	5	9	216	40 (R)	12.2	52	213	61	
D-4	4	6	199	35 (R)	24.0	27	203	59	
D-5	5	12	29	−36 (N)	2.9	719	221	53	
D-6 *	5	10	287	−60	8.4	84	132	40	
D-7	5	12	22	−36 (N)	12.5	60	213	59	
Mean and total	19	39	26	−37	7.3	159	33	−58 (N)	236
Group B									
D-4	3	4	75	52 (R)					
Diabase dykes									
Group B									
D-1	5	9	56	59 (R)	5.9	105			
D-2	5	12	68	52 (R)	18.4	19			
Mean and total	10	21	62	55					

* excluded from statistics.
N = normal; R = reverse.
N = number of samples used.
n = number of specimens used.
D, I = declination and inclination of remanent magnetization in degrees.
α_{95} = semi-angle of cone of 95% confidence for mean direction.
K = estimate of Fisher's precision parameter.

4.2. *Diabase dykes*

The stable magnetisation with coherent direction observed in diabase dyke samples of sites D1 and D2 is found in univectorial component B. The B-group remanence is a northeasterly intermediate downward directed magnetisation with intermediate T_{UB} and very low coercivity spectra.

Component B is characterised by the absence of mixed polarity (only reverse polarities are present). The group-mean remanence direction (62°, 55°) is probably not primary. There are three reasons to back up this assertion: (1) most commonly the remanence direction does not change at all even if the intensity decreases by a factor of one hundred after AF or thermal demagnetization (Fig. 7); (2) occasionally, present Earth's field is erased at intensities ranging from 0.1 to 0.01 of NRM before isolating component B (Fig. 7b); and (3) component B is seldom retrieved at lower T_{UB} and $H_{1/2}$ spectra than the PEF component (Fig. 8); this observation suggests that component B is secondary. Alternatively, its spectrum is overlapping another component and it remains unresolved. These three observations strongly point out that AF and thermal demagnetisation techniques are unable to resolve two or more superimposed components with similar magnetic spectra. The paleolatitude of this eventual component (B) is about 33°S, this value is not very different from the one of component A (39°S) and this suggests that a large fraction of this component has an ancient age of acquisition.

5. Paleomagnetic poles and their age

Based on the group-mean remanence directions, paleomagnetic poles are calculated for each group (A and B) (Table III). The age of the paleopole for

TABLE III

Paleomagnetic poles; volcanics, tuffs and diabase dykes, New Brunswick

Formation	Attitude	Component	Lat.	Paleopole Long.	d_p	d_m
Volcanics and tuffs	*In situ*	A	17°S	268°E	5°	8.1°
Volcanics and tuffs	Tilt-corrected	A	1°S	68°E	6.5°	8.8°
Volcanics and diabase dykes	*In situ*	B	33°N	10°E	–	–

the tilt corrected A component is most probably Early Devonian.

After taking into account the relative ages of the isolated components discussed in the previous section, the paleopoles for the in-situ and tilt corrected A component as well as the in-situ B component are retained in this section. The paleopole position of the *in situ* A component (in its normal polarity) is 268°E, 17°S ($d_p = 5°$, $d_m = 8.1°$). The paleopole position of the tilt-corrected A component (in its normal polarity) is 268°E, 01°S ($d_p = 6.5°$, $d_m = 8.8°$) (Fig. 9). The almost significant positive tilt correction strongly suggests that the paleopole position of the tilt-corrected A component is the one to be retained. The paleopole position of the *in situ* B component (in its normal polarity) as obtained from sites D-1, D-2 and D-4 is 10°E, 33°N ($d_p = 10.5°$, $d_m = 14.7°$).

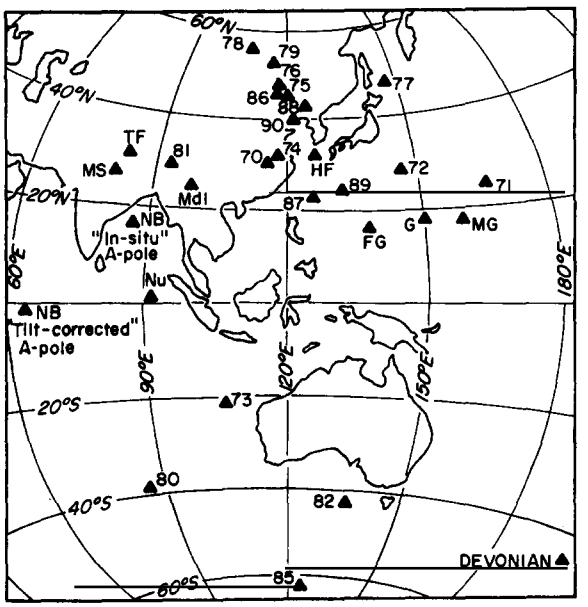


Fig. 9. Paleopoles derived from rock units of Devonian age. Redrawn from Roy et al. (1983). Additions: TF (Spariosu and Kent, 1983), NB (this study). MS, HF, FG, G and MG from Seguin et al. (1982); Mdl and Nu from Dankers (1982); numbers from Roy and Anderson (1981) and Roy et al. (1979).

First, this last paleopole is unrelated to any recent geological history because in many cases the PEF was removed in the course of AF or thermal demagnetization. Second, the B pole has an age which is younger than Middle–Upper Devonian (Acadian orogeny) since component B in site D-4 is postfolding. Third, paleopole B does not fall on any position of APW trajectory spanning the Middle-Upper Devonian to recent time interval (Irving and Irving, 1982; Roy et al., 1983). Paleopole position B (10°E, 33°N) is aberrant; the most likely interpretation is the following: Acadian and probably Hercynian orogenies overprinted the initial remanence of the low coercivity rock units. Roy and Morris (1983) demonstrated clearly that some of the poles plotted in Fig. 9 are Permo–Carboniferous overprints obtained from the Maritimes. During the Acadian orogeny (Devonian time), the paleopole was situated close to 90°E, 0° and during the Hercynian orogeny (Permian time), the paleopole was located at 125°E, 35°N. Taking a unit vectorial sum of the two components of magnetization, corresponding to the PEF (335°, 75°) and Permian time (175°, 15°), an aberrant component of magnetization B oriented in the NE quadrant and its related paleopole is to be expected.

6. Discussion and conclusion

The difficulty in determining meaningful pole positions is apparent from the results on the dykes in central New Brunswick and on the volcanics of

northern New Brunswick. The same difficulty was met in other paleomagnetic studies conducted on Devonian units from cratonic North America, the Andean margin of Iapetus and Avalon Terrane. Quite different paleomagnetic poles are obtained from rock ages which are very similar as is the case in this study (Fig. 7). The paleolatitudes of rock units of Devonian age obtained from areas not so distant are often significantly different. Given the scarcity of Devonian paleomagnetic data for North America, it is impossible to arrive at a final conclusion at this stage. However, at least for the Early Devonian interval, a coherent zone of pole positions appears to emerge from paleomagnetic studies conducted in Quebec (Seguin, 1980; Seguin et al., 1982), in Maine (Brown, 1979; Brown and Kelly, 1980; Spariosu and Kent, 1983) and more recently in New Brunswick (Roy et al., 1979, 1983 and this study) and Nova Scotia (Rao et al., 1981). In addition, the presumably primary Lower Devonian paleopole of this study agrees rather well with British poles of primary Silurian–Lower Devonian paleopoles on the pre-drift reconstruction (Briden et al., 1984). This zone of Early Devonian pole is centered around 85°E, 25°N.

New Brunswick and southern Gaspé, Québec are, to our knowledge, the best regions with the Canadian Arctic archipelago for paleomagnetic studies related to the Devonian of North America. In New Brunswick, one should continue collecting volcanic samples where tilt-corrections can be made even if that did not help very much in the Dalhousie–Campbellton area.

Even if more sites from different Devonian rock units of New Brunswick are sampled, there is no guarantee of getting the precision needed for meaningful results which can be interpreted with certainly in terms of Devonian APW path and relative movement of displaced terranes during the Acadian orogeny. This study and additional preliminary studies by one of the authors (M.K.-S) have shown that most of the Devonian basic dykes from New Brunswick are not reliable remanence indicators because of frequent overprinting. This is so because the dykes are characterized by low coercivity spectra, slowly cooling rates and relatively high degree of alteration with accompanying secondary chemical remanence.

The paleopole (268°E, 1°S) corresponding to the tilt-corrected A component of magnetisation from Lower Devonian volcanic units of New Brunswick is the same as the pole obtained from platform sediments of the same age in North America (Dankers, 1982).

The present study indicates that if the magnetisation is primary, the Lower Devonian paleomagnetic pole of northern Appalachians lies close to 90°E, equator. If the magnetization is secondary, the Middle–Upper Devonian paleomagnetic pole is located at c. 90°E, 30°N. This study suggests that by Early–Middle Devonian time, the paleopole positions for the North American platform and the northern Appalachians are not significantly different; the interpretations given to this situation is that by Early–Middle Devonian time, at least the Humber, Dunnage and Gander zones of the northern Appalachians were already juxtaposed to the North American craton.

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